ICT Tech - Class 9 (PCTB)

Class 9 — Solved Exercises

Punjab Curriculum and Textbook Board (PCTB)

6 chapters — 132 exercises

Chapter 1: Mastering Microsoft Word and PowerPoint

Multiple Choice Questions

1. What is the default file extension for Microsoft Word documents?

(a) .txt

(b) .pdf

(c) .docx 

(d) .pptx

Correct answer: .docx

Multiple Choice Questions

2. What is the function of the page break feature in Word?

(a) Deletes the current page

(b) Moves text to the next page 

(c) Copies formatting settings

(d) Aligns the document text

Correct answer: Moves text to the next page

Multiple Choice Questions

3. Which tab in PowerPoint allows users to insert images and tables?

(a) Home

(b) Insert 

(c) Design

(d) Transitions

Correct answer: Insert

Multiple Choice Questions

4. What is the primary purpose of slide master in PowerPoint?

(a) To change the background

(b) To create consistent layouts and themes of all slides ✓

(c) To add animations

(d) To modify the slideshow speed

Correct answer: To create consistent layouts and themes of all slides

Multiple Choice Questions

5. In PowerPoint, which key starts the slideshow from the beginning?

(a) F5 ✓

(b) Ctrl + S

(c) Alt + Tab

(d) Shift + Enter

Correct answer: F5

Multiple Choice Questions

6. Which file format is used to export a PowerPoint presentation as a video?

(a) .docx

(b) .mp4 ✓

(c) .pptx

(d) .xlsx

Correct answer: .mp4

Multiple Choice Questions

7. What is the default slide layout when creating a new PowerPoint presentation?

(a) Title Slide ✓

(b) Blank Slide

(c) Comparison Slide

(d) Two-Column Slide

Correct answer: Title Slide

Multiple Choice Questions

8. The spelling & grammar tool is found in which tab of Word?

(a) Insert

(b) Home

(c) Review ✓

(d) View

Correct answer: Review

Multiple Choice Questions

9. Ctrl + Z in Word and PowerPoint is used for:

(a) Saving a file

(b) Undoing an action ✓

(c) Copying text

(d) Pasting an item

Correct answer: Undoing an action

Multiple Choice Questions

10. Which tab in Word allows you to add headers and footers?

(a) Insert ✓

(b) Layout

(c) Home

(d) View

Correct answer: Insert

Short Questions

1. Define a header and a footer in Word.

A header is text or a number printed at the top of every page, and a footer is the same kind of text printed at the bottom of every page. They are used for things

that repeat on all pages, such as the document title, the page number or the date, so they do not have to be typed on each page again.

Short Questions

2. Differentiate between various types of text alignments in Word?

Left alignment lines up text at the left margin and leaves the right edge uneven; it is the usual default. Centre alignment puts each line in the middle of the page, which suits titles. Right alignment lines up text at the right margin, and Justified alignment spreads each line so both the left and right edges are straight, which suits formal paragraphs.

Short Questions

3. How can you insert an image into a PowerPoint slide?

Click on the slide, go to the Insert tab and choose Pictures, then pick the image file and click Insert. The picture appears on the slide, where it can be moved, resized by dragging its corners, and placed neatly beside the text.

Short Questions

4. Explain the difference between transitions and animations in PowerPoint.

A transition is the effect that happens when the show moves from one slide to the next, for example Fade or Push. An animation is an effect on a single object on a slide, such as a title flying in or a shape spinning. In short, transitions act on whole slides, while animations act on objects inside a slide.

Short Questions

5. How do you add bullet points to a PowerPoint slide?

Click inside the text box, go to the Home tab and click the Bullets button, then type each point on a new line. Each line then starts with a bullet dot, which

makes a list of points easy to read at a glance.

Short Questions

6. Define two ways to save a PowerPoint presentation.

First, press Ctrl + S (or click File, then Save) to save the presentation in its PowerPoint (.pptx) form so it can be edited again later. Second, use File, then Export or Save As, to save it in another form, for example as a video (.mp4) or a PDF, for sharing or showing without editing.

Short Questions

7. Explain the purpose of a Slide Master in PowerPoint.

The Slide Master holds the design that all slides follow: the background, colours, fonts and the places for titles and text. Changing the master changes every slide at once, so the whole presentation keeps one consistent layout and theme instead of each slide looking different.

Short Questions

8. Illustrate the purpose of a table in Microsoft Word.

A table arranges information in rows and columns, for example a class timetable with days across the top and periods down the side. This keeps facts neat, lined up and easy to compare, which is much clearer than writing the same facts as a long paragraph.

Long Questions

1. Explain the difference between Microsoft Word and Microsoft PowerPoint and their best use cases.

Microsoft Word is a word-processing program for writing and formatting text documents, and it is best for assignments, reports, letters and any work that is

mainly read page by page. PowerPoint is a presentation program for making slides, and it is best for showing ideas to an audience with short text, pictures, tables, charts and spoken explanation. Word suits detailed written work, while PowerPoint suits short visual points talked through in front of others.

Long Questions

2. Describe the step-by-step process of formatting a document in Word.

First, type the text, then select the part to change. Second, set the font, size and style (bold, italic or underline) from the Home tab. Third, set the alignment (left, centre, right or justified) and the line spacing so the text is easy to read. Fourth, add headings, bullet lists, page numbers and a header or footer if needed. Finally, check spelling with the Review tab and save the document with Ctrl + S.

Long Questions

3. How do SmartArt and Charts improve a PowerPoint presentation?

SmartArt turns a list or a process into a diagram, such as steps in a cycle or parts of a whole, so the audience grasps the idea at a glance instead of reading a wall of text. Charts turn numbers into bars, pies or lines, so a comparison or a trend is seen at once. Together they make the message clearer, quicker to understand and more interesting to watch.

Long Questions

4. Explain how to apply and customize slide transitions and animations.

To apply a transition, select a slide, open the Transitions tab, pick an effect such as Fade or Push, and set its duration; click Apply to All to use the same transition on every slide. To apply an animation, click the object on the slide, open the Animations tab and pick an entrance, emphasis, exit or motion-path effect, then set its timing and order in the Animation Pane. Preview the show to check the

effects, and keep them simple and smooth so the presentation still looks professional.

Long Questions

5. Discuss the importance of using consistent themes and layouts in PowerPoint presentations.

A consistent theme (the same colours, fonts and background) and the same layouts on every slide make the presentation look neat and professional, and the audience can follow the content instead of being distracted by changing designs. It also saves time, because the Slide Master sets the design once for all slides. A presentation that keeps changing style looks untidy and makes the speaker look unprepared, while a consistent one keeps attention on the message.

Chapter 2: Exploring Arduino Basics - Circuits, Code and Projects

Multiple Choice Questions

1. What is the main function of an Arduino microcontroller?

(a) Store files like a USB drive

(b) Control electronic components based on programming ✓

(c) Act as a battery for circuits

(d) Replace a computer processor

Correct answer: Control electronic components based on programming

Multiple Choice Questions

2. The buzzer in an Arduino project produces sound when:

(a) It receives an analog input

(b) It is connected to GND only

(c) A high signal (5V) is sent to its pin ✓

(d) It is placed directly on a breadboard

Correct answer: A high signal (5V) is sent to its pin

Multiple Choice Questions

3. What is the purpose of using a 220-ohm resistor in an LED circuit?

(a) Increase brightness

(b) Protect the LED from excessive current ✓

(c) Change the LED color

(d) Make the circuit faster

Correct answer: Protect the LED from excessive current

Multiple Choice Questions

4. In an Arduino sketch, which function runs only once when the program starts?

(a) setup()

(b) loop()

(c) run()

(d) start()

Correct answer: setup()

Multiple Choice Questions

5. Which tool in the Arduino IDE is used to monitor sensor data in real-time?

(a) Serial Plotter

(b) Serial Monitor

(c) Code Debugger

(d) Pin Analyzer

Correct answer: Serial Monitor

Multiple Choice Questions

6. What happens if an LED is connected backward in a circuit?

(a) It will blink twice as fast

(b) It will not turn on

(c) It will burn immediately

(d) It will work normally but at half brightness

Correct answer: It will not turn on

Multiple Choice Questions

7. What does the color "Red" mean as the third band on a four-band resistor?

(a) 2 ohms

(b) Multiply by 100 (10 to the power 2)

(c) Multiply by 1,000 (10 to the power 3)

(d) Plus or minus 5% tolerance

Correct answer: Multiply by 100 (10 to the power 2)

Multiple Choice Questions

8. A resistor has the colors Yellow, Violet, Orange, Gold. What is its resistance?

(a) 47 ohms

(b) 470 ohms

(c) 47,000 ohms (47 kilo-ohms) ✓

(d) 470,000 ohms (470 kilo-ohms)

Correct answer: 47,000 ohms (47 kilo-ohms)

Multiple Choice Questions

9. Which color represents a tolerance of plus or minus 10%?

(a) Gold

(b) Silver ✓

(c) Black

(d) Brown

Correct answer: Silver

Multiple Choice Questions

10. What is the multiplier for the color "Green" in the third band?

(a) 10 to the power 1 (10)

(b) 10 to the power 2 (100)

(c) 10 to the power 5 (100,000) ✓

(d) 10 to the power 6 (1,000,000)

Correct answer: 10 to the power 5 (100,000)

Short Questions

1. What is an Arduino Uno, and why is it commonly used in electronics projects?

The Arduino Uno is a small, low-cost microcontroller board that can be programmed to control lights, motors, buzzers and sensors. It is commonly used because it is easy to program with the Arduino IDE, it has ready digital and analog pins for connecting parts, and it lets students build and test real circuits without soldering.

Short Questions

2. What is the function of a resistor in an Arduino circuit?

A resistor limits the electric current flowing in a circuit. In an LED circuit it stops too much current from reaching the LED, which would damage or burn it; for example a 220-ohm resistor keeps the LED at a safe brightness.

Short Questions

3. Explain the difference between digital and analog pins on an Arduino board.

Digital pins work with only two states, HIGH (5V, on) or LOW (0V, off), and are used for things like LEDs and buzzers. Analog pins can read a range of changing values, such as the varying signal from a light or temperature sensor, so they suit measurements that are not simply on or off.

Short Questions

4. Decode a resistor having the following colour bands. Orange, White, Black, Silver.

Step 1: Orange is the first digit, 3. Step 2: White is the second digit, 9, giving 39. Step 3: Black is the multiplier, times 1, giving 39 x 1 = 39 ohms. Step 4: Silver is the tolerance, plus or minus 10%. So the resistor is 39 ohms, plus or minus 10%.

Short Questions

5. How does the HC-SR04 Ultrasonic Sensor measure distance?

The sensor sends out a short burst of ultrasound and waits for the echo to bounce back from an object. It measures the time the echo takes to return, and distance is worked out from that time: distance = (time x speed of sound) divided by 2, halved because the sound travels to the object and back.

Short Questions

6. What does the function pinMode(13, OUTPUT); do in an Arduino program?

It sets digital pin 13 to work as an output, so the program can send a signal out of that pin. After this, digitalWrite(13, HIGH) can switch on an LED or buzzer connected to pin 13, and digitalWrite(13, LOW) can switch it off.

Short Questions

7. Why do we use a breadboard instead of directly connecting components to the Arduino?

A breadboard lets parts be pushed in and changed quickly without soldering, so a circuit can be built, tested and corrected in minutes. Direct connections are fiddly and hard to change, while a breadboard keeps wiring tidy, lets parts be reused, and makes mistakes easy to find and fix.

Short Questions

8. What is the purpose of the Serial Monitor, and how can it help in debugging?

The Serial Monitor shows messages and sensor values sent from the Arduino to the computer in real time. When a program misbehaves, printing values there shows what the board is actually reading and doing, so a wrong reading, a loose wire or a logic mistake can be found instead of guessed at.

Long Questions

A. Labeling an Arduino Circuit Diagram: Below is a diagram of an Arduino Uno

with an LED circuit. Label the following parts: Arduino board, Breadboard, Resistor, LED, Jumper wires, Digital pin, Power (5V and GND connections). Arduino board: the blue board on the left, which runs the program. Breadboard: the white board with rows of holes on the right, which holds the circuit. LED: the small red lamp pushed into the breadboard; it lights when its pin is HIGH. Resistor: the small striped part in series with the LED, which limits its current. Jumper wires: the coloured wires linking the two boards and the parts. Digital pin: the pin header on the Arduino where a jumper wire carries the on/off signal to the LED. Power (5V and GND): the 5V and GND pins at the bottom of the Arduino, linked by the red and black wires, which supply power to the breadboard circuit.

Long Questions

B. Debugging Arduino Code (Find and Fix Errors): Below is an Arduino program that is supposed to blink an LED connected to pin 9, but it contains errors. 1. Find and correct all mistakes in the code. 2. Explain why each correction was necessary. 3. What would happen if we removed the delay(1000); lines? Corrected code: void setup() { pinMode(9, OUTPUT); } void loop() { digitalWrite(9, HIGH); delay(1000); digitalWrite(9, LOW); delay(1000); } Note printed in the notes: the book says the program contains three errors, but on the printed code only one syntax error is visible - the missing semicolon after the second delay(1000). The pin setup, the HIGH/LOW logic and the delays are all correct for an LED on pin 9, so no other change is needed; the corrected code above is the complete fix. Why the correction was necessary: every Arduino statement must end with a semicolon, so without it the program will not compile and nothing will run. If the delay(1000); lines were removed, the LED would switch on and off thousands of times a second, far too fast for the eye to see, so it would look steadily lit (and dimmer) instead of blinking once a second.

Long Questions

C. Troubleshooting a Faulty Circuit: Your classmate built an Arduino buzzer alarm circuit with an HC-SR04 sensor. However, the circuit is not working! The buzzer does not sound when an object is detected nearby, and the Serial Monitor shows incorrect distance values. The buzzer is connected to GND and digital pin 7. The sensor is connected to 5V, GND, Trig (pin 9), and Echo (pin 10). The buzzer code is correct, but the sensor data seems incorrect. 1. Identify three possible mistakes that could be causing this issue. 2. Suggest step-by-step solutions to fix the circuit. 3. Draw a corrected circuit diagram.

Three possible mistakes: (1) The Trig and Echo wires are swapped in the code or on the board, so the sensor never sends or never hears the pulse properly. (2) A loose or wrong wire on the sensor (5V, GND, Trig or Echo in the wrong row of the breadboard) or on the buzzer, so the readings are garbage. (3) The code uses the wrong pins for Trig/Echo, or the distance formula is wrong (for example the echo time is not halved), so the Serial Monitor values do not match the real distance and the buzzer condition never triggers. Step-by-step fix: first check every wire against the plan and re-seat loose jumpers; then confirm the code pins match the wired pins (Trig 9, Echo 10, buzzer 7); then print the raw echo time and the worked distance on the Serial Monitor and test at a known distance, such as 20 cm; finally check the buzzer alone with a simple HIGH/LOW test. Corrected circuit diagram (described): Arduino 5V to sensor VCC and GND to sensor GND; Arduino pin 9 to sensor Trig and pin 10 to sensor Echo; Arduino pin 7 to the buzzer positive and buzzer negative to GND; all parts sharing the same GND rail on the breadboard.

Chapter 3: Graphic Designing and Multimedia

Multiple-Choice Questions

1. What is graphic design?

(a) Writing stories for movies

(b) Building websites only

(c) Making things look good using pictures, colours, and text ✓

(d) Recording videos for school

Correct answer: Making things look good using pictures, colours, and text

Multiple-Choice Questions

2. What is the main difference between digital and print publishing?

(a) Digital is on paper, print is online

(b) Digital is on a screen, print is on paper ✓

(c) Digital uses text, print uses videos

(d) Digital is for posters, print is for videos

Correct answer: Digital is on a screen, print is on paper

Multiple-Choice Questions

3. Which tool is best for creating a poster in Canva?

(a) Whiteboard

(b) Video

(c) Poster ✓

(d) Website

Correct answer: Poster

Multiple-Choice Questions

4. What does a QR code do when added to a poster?

(a) It prints the poster faster

(b) It links to a website, video, or file when scanned ✓

(c) It changes the poster's colour

(d) It tracks how many posters are printed

Correct answer: It links to a website, video, or file when scanned

Multiple-Choice Questions

5. How does CapCut differ from Canva, as per the chapter?

(a) CapCut is only for printing, Canva is for videos

(b) CapCut focuses on video editing, Canva is better for static designs ✓

(c) CapCut is for posters, Canva is for barcodes

(d) CapCut has no animations, Canva does

Correct answer: CapCut focuses on video editing, Canva is better for static designs

Multiple-Choice Questions

6. Which of the following is NOT a feature of Canva?

(a) Creating presentations

(b) Editing videos

(c) Writing programming code ✓

(d) Designing social media posts

Correct answer: Writing programming code

Multiple-Choice Questions

7. What is the main purpose of the "Elements" tool in Canva?

(a) To add text to designs

(b) To insert shapes, icons, and illustrations ✓

(c) To export designs as PDFs

(d) To adjust image brightness

Correct answer: To insert shapes, icons, and illustrations

Multiple-Choice Questions

8. Which of the following features is available in CapCut?

(a) Code debugging

(b) 3-D animation modeling

(c) Video trimming and transitions 

(d) Web hosting

Correct answer: Video trimming and transitions

Multiple-Choice Questions

9. What is the main difference between Canva and CapCut?

(a) Canva is for video editing, while CapCut is for graphic design

(b) Canva is used for graphic design, while CapCut is used for video editing 

(c) Both are used for the same purpose

(d) Canva is a mobile app, but CapCut is only for desktops

Correct answer: Canva is used for graphic design, while CapCut is used for video editing

Multiple-Choice Questions

10. Which of the following is true about Canva and CapCut?

(a) Both tools allow advanced coding features

(b) Canva is best for video transitions, while CapCut is best for posters

(c) Canva is mainly for designing images and presentations, while CapCut specializes in editing videos 

(d) CapCut can create animations, but Canva cannot

Correct answer: Canva is mainly for designing images and presentations, while CapCut specializes in editing videos

Short Questions

1. What is the difference between digital publishing and print publishing? Give one example of each.

Digital publishing shares a design on a screen, for example posting a poster on the school website or social media, where it can be sent to friends with a click. Print publishing makes physical copies, for example printing flyers for a school event to hand out. Digital suits videos and online sharing; print suits handing a copy to someone in person.

Short Questions

2. Name two key design basics you should follow when making a poster.

First, Colours: use two or three bright colours to grab attention, but do not mix too many, for example red and yellow for a fun event. Second, Balance and Contrast: spread things out so the poster is not crowded, and use dark text on a light background (or the reverse) so the message stands out and is easy to read.

Short Questions

3. List three types of designs you can create with Canva.

Posters for school events, presentations for class, and social media posts are three designs Canva is made for. Canva also offers flyers, brochures and other ready templates, all built from its text, elements and picture tools.

Short Questions

4. Why is CapCut good for creating a video flyer for a school event?

CapCut is a video-editing tool, so it can trim clips, add transitions, music and text, and join short scenes into one lively flyer video. A moving video catches attention far better than a still notice, which is exactly what a school event announcement needs.

Short Questions

5. What is a documentation system, and why is it important for managing your designs?

A documentation system is an organized way of saving and naming design files, for example keeping a project folder with the poster, the video and a list of what was given to whom. It matters because it stops files being lost or mixed up, makes the latest version easy to find, and lets the team track printed copies and online links.

Long Questions

1. Explain the steps to create a poster for a school event using Canva. Include at least five steps in your answer.

Step 1: Open Canva and choose the Poster template size. Step 2: Pick a background colour or picture that suits the event. Step 3: Add the event title in a big, easy-to-read font, with the date, time and place in smaller text. Step 4: Use Elements to add shapes, icons or pictures that match the theme, keeping two or three colours and a balanced layout. Step 5: Check the spelling and spacing, then download the poster as a PDF for printing or as an image for sharing online.

Long Questions

2. Describe how to make a QR code for a school project and add it to a poster. Why is this useful?

First, save the project file with a clear name, such as TalentShowPoster_Final.pdf, and upload it to Google Drive. Next, create a QR code for that Drive link using a QR generator. Then place the QR code on the printed poster where it is easy to scan. It is useful because anyone with a phone can scan the poster and see more details online at once, instead of the poster having to carry every detail in print.

Long Questions

3. Compare Canva and CapCut as design tools. Give two differences and explain how you would use each for a school talent show project.

First difference: Canva is mainly for still designs such as posters, presentations and social media images, while CapCut specializes in editing videos. Second difference: Canva builds a design from templates, text and elements, while CapCut builds a video from clips, trimming, transitions and sound. For a talent show, Canva would make the poster and the printed programme, and CapCut would edit the short announcement video and the highlights video afterwards.

Long Questions

4. What are barcodes, and how can you use them to manage printed flyers for a school event? Describe the steps to create a barcode.

A barcode is a printed pattern of bars that a scanner can read to identify an item or a batch. For flyers, a barcode on the stack lets the team track how many copies are left. Steps: first, print the flyers and stack them neatly; second, generate a barcode for the batch (for example one code per class set); third, stick or print the barcode on the stack; fourth, scan it whenever copies are given out, and keep a written list, such as 10 to Class 5 and 5 to Teachers, so the remaining count is always known.

Long Questions

5. Why is it important to organize your digital and print media? Describe how you would manage a project with a poster and video for a school fair, including QR codes and barcodes.

Organizing media matters because a lost file, a mixed-up version or an uncounted stack wastes time and can spoil the event publicity. For a school fair: save the poster and video in one named folder, with the final versions clearly

marked. Upload the video to Drive and put its QR code on the printed poster so visitors can watch it by scanning. Print the posters, stack them neatly, add a barcode to each stack, and keep a list of who received copies. This way the digital files are easy to find, the printed copies are counted, and nothing is handed out twice or lost.

Chapter 4: Fundamentals of Web Development and e-Commerce

Multiple Choice Questions

1. Which language provides the structure for a webpage?

(a) CSS

(b) JavaScript

(c) HTML ✓

(d) Python

Correct answer: HTML

Multiple Choice Questions

2. Which attribute is used to specify an image source in an <img> tag?

(a) Src ✓

(b) href

(c) Alt

(d) source

Correct answer: Src

Multiple Choice Questions

3. Choose the purpose of the <form> element in HTML

(a) To structure text content

(b) To define a section in a webpage

(c) To collect user input ✓

(d) To create a navigation bar

Correct answer: To collect user input

Multiple Choice Questions

4. Which tag is used to create a hyperlink in an HTML document?

(a) <link>

(b) <a> ✓

(c) <href>

(d) <ur>

Correct answer: <a>

Multiple Choice Questions

5. What does CSS stand for?

(a) Computer Styling Sheets

(b) Cascading Style Sheets ✓

(c) Creative Style Syntax

(d) Common Styling System

Correct answer: Cascading Style Sheets

Multiple Choice Questions

6. Which combination of technologies is commonly used for front-end web development?

(a) JavaScript, CSS, PHP

(b) HTML, CSS, JavaScript ✓

(c) HTML, Python, SQL

(d) CSS, Java, C++

Correct answer: HTML, CSS, JavaScript

Multiple Choice Questions

7. Which JavaScript function is used to display messages in a pop-up box?

(a) console.log()

(b) alert() ✓

(c) print()

(d) messageBox()

Correct answer: alert()

Multiple Choice Questions

8. Which symbol is used for single-line comments in JavaScript?

(a) /* Comment */

(b) // Comment 

(c) <! Comment -->

(d) # Comment

Correct answer: // Comment

Multiple Choice Questions

9. What is e-Commerce, as explained in the chapter?

(a) Building websites for schools

(b) Buying and selling things on the internet 

(c) Making videos for shops

(d) Writing stories online

Correct answer: Buying and selling things on the internet

Multiple Choice Questions

10. What is the first step to set up an online store on Daraz?

(a) Add a product photo

(b) Sign up with your email 

(c) Ship the product

(d) Write a description

Correct answer: Sign up with your email

Short Questions

1. What does the <p> HTML tag do on a webpage?

The <p> tag marks a paragraph of text on a webpage. The browser shows its content as a block of text with a gap before and after it, so separate paragraphs stay clearly apart instead of running together.

Short Questions

2. Write an HTML tag to display an image from a file named "photo.jpg".

The tag is: <img src="photo.jpg" alt="Photo">. The src attribute gives the file name (the image source), and the alt attribute gives short text that shows if the picture cannot load.

Short Questions

3. What does <a> tag do on a webpage?

The <a> (anchor) tag creates a hyperlink. Its href attribute gives the address to jump to, so clicking the linked text or picture takes the visitor to another page, section or website.

Short Questions

4. Write a simple CSS rule to change the background color of a webpage to green.

The rule is: body { background-color: green; }. The selector body picks the whole page, and the background-color property sets its background to green.

Short Questions

5. How can CSS classes be used to style multiple elements at once?

A class is defined once in CSS, for example .note { color: blue; }, and then added to any number of elements with class="note". Every element carrying that class gets the same style at once, so changing the one rule updates them all.

Short Questions

6. What does the JavaScript function getElementById() do?

It finds one element on the page by its id, so a script can work with that exact element. For example, getElementById("title") returns the heading with id="title", whose text or style the script can then change when a button is clicked.

Short Questions

7. What is e-Commerce? Give one example of an e-Commerce platform in Pakistan.

E-Commerce is buying and selling things on the internet, instead of in a physical shop. Daraz is a well-known e-Commerce platform in Pakistan, where sellers list products online and buyers order and pay for them through the website or app.

Short Questions

8. Why is it important to talk to the buyer after they place an order?

Talking to the buyer confirms the order, clears up any doubt about size, colour or delivery, and builds trust. A buyer who is kept informed is far more likely to be satisfied, to accept the delivery happily, and to buy again or recommend the shop.

Long Questions

1. Write a simple HTML code to create a webpage with a heading that says "My Online Shop" and a paragraph that says "Welcome to my store!" Then, add CSS to make the heading green and the paragraph text blue.

HTML: <html> <head> <title>My Online Shop</title> <style> h1 { color: green; } p { color: blue; } </style> </head> <body> <h1>My Online Shop</h1> Welcome to my store! </body> </html>. The h1 rule

colours the heading green, the p rule colours the paragraph blue, and the two text lines appear exactly as asked.

Long Questions

2. Create a JavaScript function that changes the text inside an <h1> tag when a button is clicked.

Give the heading an id, for example <h1 id="title">Old Text</h1>, and add a button <button onclick="changeText()">Change</button>. Then write: function changeText() { document.getElementById("title").innerHTML = "New Text"; }. When the button is clicked, the function finds the heading by its id and replaces its text with the new text.

Long Questions

3. Make a simple calculator using JavaScript that allows users to enter two numbers and choose an operation (addition, subtraction, multiplication, or division). Finally display the result.

First, make two number inputs and a dropdown for the operation (+, -, x, /), plus a button and a place to show the answer. On clicking the button, read both numbers with getElementById, check which operation was chosen, and work it out: addition adds, subtraction subtracts, multiplication multiplies, and division divides (checking that the second number is not zero, since division by zero is not allowed). Finally, show the result in the answer area, for example Result: 15.

Long Questions

4. Why is it important to manage payments carefully in an online store? Describe how you would keep track of orders and payments for a shop selling notebooks, including one HTML tag you could use to display the shop's name on a webpage.

Careful payment management matters because mistakes lose money, upset

buyers and can close a small shop: an unpaid order shipped by mistake, or a paid order forgotten, both damage trust. For a notebook shop: set up payment options (bank card or cash on delivery) on Daraz or Shopify; after each order, check that the payment is done (for example a note saying Payment Received: Rs. 400); and write down every payment, such as Ayesha paid Rs. 400 on 10 March, in a notebook or sheet. The shop name on a webpage can be shown with the heading tag, for example <h1>Notebook Shop</h1>.

Long Questions

5. Explain the steps to set up an online store on Daraz to sell handmade keychains. Include at least four steps.

Step 1: Sign up with your email to create a seller account on Daraz. Step 2: Add your keychains as products, with a clear photo of each design. Step 3: Write a description and a fair price for each keychain. Step 4: Set up payment options, such as bank card or cash on delivery, so buyers can pay safely. Step 5: When an order comes, pack and ship the keychain, keep the buyer informed, and record the payment, for example Keychain order paid Rs. 400.

Chapter 5: System Troubleshooting and Maintenance

Multiple Choice Questions

1. Which tool is used to remove unnecessary files from a hard drive?

(a) Disk Defragmenter

(b) Task Manager

(c) Disk Cleanup ✓

(d) Device Manager

Correct answer: Disk Cleanup

Multiple Choice Questions

2. What is the main purpose of disk defragmentation?

(a) To delete unused files

(b) To organize fragmented files for faster access ✓

(c) To increase disk space

(d) To clean viruses

Correct answer: To organize fragmented files for faster access

Multiple Choice Questions

3. Which of the following drives should NOT be defragmented?

(a) HDD (Hard Disk Drive)

(b) External USB Drive

(c) SSD (Solid State Drive) ✓

(d) Network Drive

Correct answer: SSD (Solid State Drive)

Multiple Choice Questions

4. How do you check if a drive needs defragmentation in Windows?

(a) Open Task Manager

(b) Run Disk Cleanup

(c) Use the "Analyze" option in Disk Defragmenter ✓

(d) Restart the computer

Correct answer: Use the "Analyze" option in Disk Defragmenter

Multiple Choice Questions

5. Which Windows tool helps boot the system with minimal drivers for troubleshooting?

(a) Device Manager

(b) Safe Mode ✓

(c) Disk Cleanup

(d) Task Scheduler

Correct answer: Safe Mode

Multiple Choice Questions

6. What should you do if a program is causing system crashes?

(a) Update the operating system

(b) Reinstall the program

(c) Uninstall the program ✓

(d) Change the system language

Correct answer: Uninstall the program

Multiple Choice Questions

7. If Safe Mode does not fix the issue, what is the next step?

(a) Use Windows Recovery Environment (WinRE) ✓

(b) Install more RAM

(c) Unplug the monitor

(d) Run Disk Defragmenter

Correct answer: Use Windows Recovery Environment (WinRE)

Multiple Choice Questions

8. What is the main function of antivirus software?

(a) Speed up the internet

(b) Scan and remove malicious programs 

(c) Install new software

(d) Improve screen resolution

Correct answer: Scan and remove malicious programs

Multiple Choice Questions

9. What is "phishing"?

(a) A type of computer game

(b) A method of stealing personal information through fake emails 

(c) A tool for speeding up a computer

(d) A type of software update

Correct answer: A method of stealing personal information through fake emails

Multiple Choice Questions

10. What does an IP address do?

(a) Stores files on a computer

(b) Identifies devices on a network 

(c) Increases internet speed

(d) Detects malware

Correct answer: Identifies devices on a network

Short Questions

1. Name two system tools used for computer maintenance.

Disk Cleanup, which removes unnecessary and temporary files from a hard drive, and Disk Defragmenter, which reorganizes fragmented files for faster access. Other maintenance tools in the chapter are System Restore, Safe Mode and Windows Recovery Environment (WinRE).

Short Questions

2. What does Disk Defragmenter do?

Over time a hard drive stores parts of one file in scattered places, which slows reading. Disk Defragmenter gathers those scattered parts together in order, so the drive reads each file in one smooth pass and the computer runs faster.

Short Questions

3. Why should an SSD not be defragmented?

An SSD has no moving parts and reads any stored part equally fast, so gathering fragments together gives no speed gain. Defragmenting only adds useless write cycles, which wear the drive out sooner, so SSDs must not be defragmented.

Short Questions

4. What is the purpose of a System Restore?

System Restore returns the computer's system files and settings to an earlier point in time, called a restore point, when the computer was working well. It is used to undo a bad update, a faulty driver or a wrong setting without deleting the user's personal files.

Short Questions

5. How can malware be prevented?

Install trusted antivirus software and keep it updated so it can scan and remove malicious programs. Also avoid opening unknown email attachments or links,

which is how phishing and other malware spread, and keep the operating system updated.

Short Questions

6. What is the difference between a static and dynamic IP address?

A static IP address is fixed and never changes, so a device is always found at the same address, which suits servers and hosted services. A dynamic IP address is assigned fresh each time a device joins the network and may change later, which is the usual setup for home computers and phones.

Short Questions

7. What command is used to check the IP address of a computer?

On Windows, open the Command Prompt and type ipconfig, then press Enter. The result shows the computer's IP address, subnet mask and default gateway for its network connection.

Short Questions

8. What is port forwarding?

Port forwarding is a router setting that sends incoming traffic on a chosen port straight to one chosen device on the local network. It is useful for gaming, remote access, viewing security cameras remotely, and hosting websites or servers, because outsiders can then reach that one device through the router.

Long Questions

1. Explain the importance of computer maintenance and list some key maintenance tasks.

Regular maintenance keeps a computer fast, reliable and safe, and it prevents small problems, such as a full disk or an outdated antivirus, from turning into

crashes or data loss. Key tasks are: removing unnecessary files with Disk Cleanup; defragmenting a hard drive (never an SSD) with Disk Defragmenter; updating the operating system and antivirus; checking the drive with the Analyze option; and making restore points so a fault can be undone with System Restore.

Long Questions

2. Describe the role of Disk Defragmenter and explain the process of defragmenting a hard drive.

Disk Defragmenter reorganizes fragmented files on a hard drive so each file's parts sit together, which makes reading faster. Process: first open Disk Defragmenter and select the drive; second, click Analyze to check whether the drive actually needs defragmentation; third, click Optimize (Defragment) and let it run without switching the computer off; fourth, check the result and repeat only when analysis shows real fragmentation. SSDs are excluded from this process entirely.

Long Questions

3. What are common software troubleshooting methods used when a computer fails to start properly?

First, boot into Safe Mode, which starts Windows with minimal drivers, so a faulty driver or program can be found. If Safe Mode does not fix it, use Windows Recovery Environment (WinRE) to repair startup, or use System Restore to return to a restore point from when the computer worked. If one program is causing crashes, reinstall it, and if it still crashes, uninstall it. Keeping notes of what changed just before the fault helps pinpoint the cause.

Long Questions

4. What are the key differences between static and dynamic IP addresses? Provide their advantages and disadvantages.

A static IP never changes, which is an advantage for servers, cameras and remote access, since the device is always at the same address; its disadvantage is that it must be set up by hand and can be easier to target. A dynamic IP is handed out automatically and may change, which is an advantage because it needs no setup and suits most home users; its disadvantage is that the address cannot be relied on for hosting, since it may change at any time.

Long Questions

5. Explain port forwarding and its applications in networking.

Port forwarding tells a router to pass traffic arriving on a certain port to one specific device inside the local network, instead of blocking it. Its applications, given in the chapter, are: multiplayer games connecting smoothly; remote access, so a user can control a home computer from anywhere; viewing live security-camera footage remotely; and letting others access hosted content, such as a website or server running on a home computer.

Chapter 6: AI Technologies

Multiple Choice Questions

1. Which AI tool is commonly used for grammar and spelling correction?

(a) ChatGPT

(b) Grammarly ✓

(c) QuillBot

(d) DALL.E

Correct answer: Grammarly

Multiple Choice Questions

2. What is the main purpose of AI summarization tools?

(a) To write original content

(b) To shorten long texts while retaining key points ✓

(c) To translate languages

(d) To create images

Correct answer: To shorten long texts while retaining key points

Multiple Choice Questions

3. Which AI tool can generate images from text descriptions?

(a) Canva

(b) DALL.E ✓

(c) QuillBot

(d) Zapier

Correct answer: DALL.E

Multiple Choice Questions

4. What does AI use to recognize objects in images?

(a) Natural Language Processing (NLP)

(b) Computer Vision ✓

(c) Speech Recognition

(d) Predictive Analytics

Correct answer: Computer Vision

Multiple Choice Questions

5. Which AI tool helps developers write code faster?

(a) GitHub Copilot ✓

(b) Grammarly

(c) Canva

(d) Zapier

Correct answer: GitHub Copilot

Multiple Choice Questions

6. Which industry uses AI for personalized recommendations?

(a) Healthcare

(b) Retail ✓

(c) Education

(d) Transportation

Correct answer: Retail

Multiple Choice Questions

7. What is the primary use of AI in healthcare?

(a) To play games

(b) To diagnose diseases and analyze medical images ✓

(c) To write novels

(d) To create music

Correct answer: To diagnose diseases and analyze medical images

Multiple Choice Questions

8. What is the main ethical concern of AI in daily life?

(a) Job displacement

(b) Privacy

(c) Bias

(d) All of these ✓

Correct answer: All of these

Multiple Choice Questions

9. What is bias in AI?

(a) A type of AI algorithm

(b) Unfair or discriminatory outcomes in AI systems ✓

(c) A programming language

(d) A type of data storage

Correct answer: Unfair or discriminatory outcomes in AI systems

Multiple Choice Questions

10. Which of the following is a privacy concern related to AI?

(a) Data breaches ✓

(b) Bias in algorithms

(c) Job displacement

(d) Lack of creativity

Correct answer: Data breaches

Short Answer Questions

1. How does AI help in Speech-to-text generation?

AI speech recognition listens to spoken words, breaks the sound into patterns, and matches those patterns to words and sentences. This turns speech into

written text, which is used in dictation, subtitles, voice assistants and meeting notes.

Short Answer Questions

2. How does AI help in video editing?

AI video tools can trim clips, add transitions, pick highlights and suggest music automatically, which used to take a long editing session by hand. This lets a student or creator finish a clean, well-paced video far more quickly.

Short Answer Questions

3. What is the purpose of AI in image creation?

AI image tools, such as DALL.E, create pictures from a written description, so a user can get an original illustration without drawing or photography. This is useful for posters, presentations and design ideas when no ready picture exists.

Short Answer Questions

4. What is GitHub Copilot used for?

GitHub Copilot is an AI coding assistant that suggests lines or whole blocks of code as a developer types. It helps developers write code faster and with fewer routine mistakes, while the developer still checks and corrects the suggestions.

Short Answer Questions

5. What is the main benefit of AI in task automation?

The main benefit is that repetitive, routine work is done quickly and without tiring, which saves time and cuts human error. People are then free for creative work, decisions and tasks that need judgement, while the machine handles the repeated steps.

Short Answer Questions

7. Give an example of AI in daily life.

A voice assistant on a phone, such as one that answers questions, sets reminders or plays music on a spoken command, is AI in daily life. Other everyday examples are personalized recommendations on shopping or video apps, and grammar correction while typing.

Short Answer Questions

8. How is AI used in the retail industry?

Retail uses AI for personalized recommendations, suggesting products a customer is likely to want based on past browsing and buying. It also helps with stock control, demand prediction and chatbots that answer shoppers' questions at any hour.

Long Answer Questions & Practical Challenges

1. Explain how AI is used in healthcare. Provide examples.

AI in healthcare mainly diagnoses diseases and analyzes medical images, for example scanning X-rays or scans to spot signs of illness earlier than the eye alone might. It also helps predict patient risk, plan treatment, and manage hospital records, and chatbots can answer basic health questions. These tools support doctors and speed up care, but the final medical decision stays with a trained professional.

Long Answer Questions & Practical Challenges

2. What are the advantages and limitations of AI in Education?

Advantages: AI gives instant feedback, personalizes practice to each student's level, summarizes long texts while keeping key points, and corrects grammar and spelling. Limitations: it can be biased, it can make confident mistakes, it

raises privacy questions about student data, and over-reliance on it can weaken a student's own thinking and writing. Used as a helper with teacher guidance, it strengthens learning; used as a replacement for thinking, it weakens it.

Long Answer Questions & Practical Challenges

3. Explain how AI is used in the transportation industry.

Transportation uses AI for route planning and traffic prediction, so drivers and delivery fleets avoid jams and save fuel. It powers driver-assistance and self-driving systems, which use computer vision to recognize roads, vehicles and pedestrians. It also helps airlines and railways predict demand and manage schedules, making travel safer and more efficient.

Long Answer Questions & Practical Challenges

4. What are the key features of an AI-powered recommendation system?

A recommendation system learns from a user's past choices, searches and ratings, and from the choices of similar users. Its key features are personalization (different suggestions for different people), continuous learning (suggestions improve as more data arrives), and ranking (the most relevant items appear first). Retail shops and video apps use it to show products or videos each person is most likely to enjoy.

Long Answer Questions & Practical Challenges

5. How can AI be used to address environmental challenges?

AI can analyze large environmental data, such as satellite images and weather records, to track deforestation, pollution and climate change. It helps predict floods, wildfires and crop needs, optimize energy use in buildings and grids, and plan efficient transport to cut emissions. In short, it turns huge data into early warnings and smarter use of resources.